Progressive Microcephaly–Seizures–Cortical Blindness–Developmental Delay Syndrome (SCBMS): A DIAPH1 Cytoskeletopathy

Disease: Progressive Microcephaly–Seizures–Cortical Blindness–Developmental Delay Syndrome (SCBMS) **MONDO:** MONDO:0014714 · **OMIM:** 616632 · **Causal gene: DIAPH1** (HGNC:2876; protein product **mDia1**) · **Category:** Mendelian, autosomal recessive

Summary

Progressive Microcephaly–Seizures–Cortical Blindness–Developmental Delay Syndrome (**SCBMS**) is an ultra-rare autosomal-recessive neurodevelopmental and neuro-immunological disorder caused by **biallelic loss-of-function (LOF) variants in *DIAPH1***, the gene encoding the RhoA-effector formin **mDia1** (Diaphanous-related formin 1). The disorder was first established as a human Mendelian condition by Ercan-Sencicek et al. in 2015 (PMID: 24781755) and has since been delineated across several cohorts, culminating in a 53-individual series in 2026 (PMID: 41860019). It must be carefully distinguished from the *allelic* but mechanistically opposite dominant **gain-of-function** DIAPH1 disorder **DFNA1** (non-syndromic sensorineural deafness with macrothrombocytopenia).

The core clinical phenotype comprises **early-infantile-onset (mean ~2.3 months) progressive postnatal microcephaly, epilepsy, cortical visual impairment/blindness, and severe developmental delay/intellectual disability**, frequently accompanied by occipital-predominant structural brain abnormalities. Beyond the neurological core, DIAPH1 loss produces a striking multisystem signature: **combined immunodeficiency** (low T-cell receptor excision circles [TRECs], low naïve T cells, and T/NK/ILC defects), **mitochondrial dysfunction**, a **DNA-repair defect**, and a notable **predisposition to B-cell lymphoma** (3 of 7 patients in one cohort).

Mechanistically, all of these features converge on the loss of mDia1's dual cytoskeletal function: as a RhoA effector, mDia1 elongates **long, linear, unbranched actin filaments** and coordinates/stabilizes **microtubules**, functions essential for mitosis, cell polarity, migration, and microtubule-organizing-center (MTOC) positioning in both **neuroepithelial precursors / cortical interneurons** and **lymphocytes**. The result is a unified "**cytoskeletopathy**" in which a single molecular lesion branches into neurodevelopmental, immune, and genome-stability phenotypes. No curative therapy exists; diagnosis is molecular (exome/genome sequencing), management is supportive and multidisciplinary, and the prognosis is guarded.

Key Findings

Finding 1 — Biallelic LOF of *DIAPH1* causes SCBMS (autosomal recessive)

SCBMS (MONDO:0014714; OMIM 616632) is caused by **biallelic loss-of-function variants in *DIAPH1***, which encodes the formin mDia1. Multiple independent cohorts converge on this conclusion:

- **Ercan-Sencicek et al., 2015** (PMID: 24781755) — founding study; identified a homozygous nonsense variant **p.Gln778*** by linkage plus whole-exome sequencing (WES) in a consanguineous family with microcephaly, severe visual impairment, intellectual disability, and short stature.
- **Kaustio et al., 2021** (PMID: 33662367) — 7 patients; 5 Finnish homozygous for the splice variant **c.684+1G>A**, 2 Omani homozygous for the frameshift **c.2769delT / p.F923fs**.
- **Esmaeilzadeh et al., 2024** (PMID: 39578953) — novel homozygous nonsense **c.1285C>T / p.Gln429*** plus review of 20 prior cases; recurrent variants **c.2769del, c.684+1G>A, c.2332C>T** account for ~72% of cases.
- **Galassi Deforie et al., 2026** (PMID: 41860019) — largest cohort, 53 individuals (33 newly reported).

> "Homozygous loss of DIAPH1 results in seizures, cortical blindness, and microcephaly syndrome (SCBMS)." — PMID: 33662367
>
> "Biallelic DIAPH1 pathogenic variants cause a neurodevelopmental syndrome occasionally associated with immunodeficiency." — PMID: 41860019

The inheritance is **autosomal recessive**, with nonsense, frameshift, and splice LOF variants. This contrasts sharply with the dominant **gain-of-function** DIAPH1 variants that cause DFNA1 / macrothrombocytopenia:

> "Dominant gain-of-function DIAPH1 variants cause macrothrombocytopenia and sensorineural deafness (autosomal dominant non-syndromic hearing loss 1 (DFNA1)), while homozygous loss of DIAPH1 results in seizures, cortical blindness, and microcephaly syndrome (SCBMS)." — PMID: 38915998

Key identifiers: MONDO:0014714 · OMIM:616632 · gene DIAPH1 (HGNC:2876, chromosome 5q31.3). Synonyms: *seizures, cortical blindness, microcephaly syndrome*; SCBMS. Information is derived from aggregated disease-level resources (OMIM/Orphanet) and from individual-patient case series/cohorts in the primary literature.

Finding 2 — Core clinical phenotype: progressive microcephaly, epilepsy, cortical visual impairment, developmental delay, early infantile onset

The defining tetrad of SCBMS is **progressive (postnatal) microcephaly, epilepsy, cortical visual impairment/blindness, and severe developmental delay/intellectual disability**, with onset in early infancy.

> "Clinical features included developmental delay, intellectual disability, progressive microcephaly, cortical visual impairment or blindness, epilepsy, and frequent occipital-predominant brain abnormalities." — PMID: 41860019 (n=53)
>
> "The mean age of onset for patients in the group of 20 patients with a known age of onset was 2.3 months (SD = 1.6)." — PMID: 39578953

Nearly half of patients in the large cohort had infections (mainly respiratory, related to epilepsy/aspiration). The 2021 cohort additionally documented mitochondrial dysfunction and immunodeficiency in the same patients (PMID: 33662367). The phenotype is **progressive** (microcephaly worsens postnatally) and **severe**, with substantial quality-of-life impact: profound disability, dependence for daily activities, visual loss, and seizure burden.

Suggested HPO terms: Progressive microcephaly (HP:0000253), Seizure (HP:0001250), Cortical visual impairment (HP:0100704), Global developmental delay (HP:0001263), Intellectual disability (HP:0001249), Cerebral atrophy (HP:0002059), Cerebellar atrophy (HP:0001272), Recurrent respiratory infections (HP:0002205), Short stature (HP:0004322).

Finding 3 — Multisystem signature: combined immunodeficiency, mitochondrial dysfunction, DNA-repair defect, and B-cell lymphoma risk

DIAPH1 loss extends well beyond the brain. Kaustio et al. (PMID: 33662367) found poor lymphocyte activation/proliferation, defective B-cell maturation, absence of naïve T cells, and mitochondrial dysfunction; critically, **CRISPR DIAPH1 knockout in healthy-donor PBMCs replicated the T-cell activation defect**, and patient T cells showed impaired adhesion and inefficient MTOC repositioning. Three of seven patients developed **B-cell lymphoma**.

> "the patients presented with infection susceptibility due to defective lymphocyte maturation and 3 patients developed B-cell lymphoma" — PMID: 33662367

Azizoglu et al., 2024 (PMID: 39120629) reported six patients with novel LOF variants and major **T, NK, and ILC defects**:

> "Loss of function mutations in Diaphanous related formin 1 (DIAPH1) are associated with seizures, cortical blindness, and microcephaly syndrome (SCBMS) and are recently linked to combined immunodeficiency." — PMID: 39120629

The large 2026 cohort confirmed a consistent immunological laboratory signature even when gross subsets looked normal:

> "T cell receptor excision circles and naïve T-lymphocyte counts were consistently low" — PMID: 41860019

Finding 4 — Mechanism: mDia1 is a RhoA-effector formin building linear actin and stabilizing microtubules

DIAPH1/mDia1 is a **Diaphanous-related formin** and a canonical downstream effector of the small GTPase **RhoA**. When RhoA-GTP binds mDia1, it relieves the protein's N-terminal (DID)–C-terminal (DAD) autoinhibition, activating the **FH2 domain** to nucleate and processively elongate long, linear, unbranched actin filaments that resist myosin-driven contraction; mDia1 also coordinates and stabilizes microtubules.

> "mDia1 generates linear actin filaments in the maintenance of polarity during adhesion, migration, and division in immune cells and neuroepithelial cells, and in driving tangential migration of cortical interneurons in the rodent" — PMID: 24781755

Supporting mechanistic cell biology: - mDia1 regulates **microtubule stability and dynamics** as a Rho effector coordinating actin and MT cytoskeletons (PMID: 25298036). - The **Memo–RhoA–mDia1** axis controls MT dynamics, adhesion-site formation, and migration (PMID: 18955552). - mDia-family formins assemble long linear filaments capable of withstanding myosin-driven contractile load; combined formin loss causes reduced cortical rigidity, excessive blebbing, and delayed cytokinesis (PMID: 42827260). - Truncation of the FH2 domain (p.R1049X) abolishes F-actin elongation activity and impairs cell–cell junction formation and mitosis (PMID: 35060117).

Because mDia1 protein is expressed in human neuronal precursor cells during mitotic division, loss impairs the mitosis, polarity, and migration of neural precursors and cortical interneurons — the cell-biological root of microcephaly and cortical malformation.

Suggested GO BP terms: actin filament polymerization (GO:0030041), actin nucleation (GO:0045010), regulation of microtubule cytoskeleton organization (GO:0070507), neuron migration (GO:0001764), mitotic cell cycle (GO:0000278), establishment of cell polarity (GO:0030010).

Finding 5 — Model organisms partially recapitulate the human disease

Model	Genetic manipulation	Phenotype recapitulation	Key limitation
Mouse (*Diap1* KO)	Constitutive knockout	Grossly **normal body and brain size**; unilateral ventricular enlargement on histology	**Does not model human microcephaly**
Xenopus tropicalis	Loss of function	**Mirrors growth and eye defects** seen in humans	Non-mammalian
Zebrafish	Loss of function	No overt malformation; **seizure-like behavior** in Phenothiazine assays	Does not reproduce structural brain defects
Human PBMC (CRISPR KO)	In-vitro DIAPH1 KO	**Replicates T-cell activation defect**	Immune phenotype only

> "diap1 (mDia1 knockout (KO))-deficient mice have grossly normal body and brain size" — PMID: 24781755
>
> "The Xenopus model mirrored growth and eye defects seen in humans, whereas zebrafish exhibited no overt malformations but showed seizure-like behavior" — PMID: 41860019

The mouse KO's lack of microcephaly is an important caveat — it likely reflects **functional redundancy** among mammalian formins (e.g., mDia2/mDia3) and species differences in cortical neurogenesis. The human in-vitro CRISPR PBMC model (PMID: 33662367) is the strongest direct causal evidence for the immune phenotype.

Finding 6 — Epidemiology and inheritance: ultra-rare, consanguinity-enriched, with a Finnish founder allele

SCBMS is **ultra-rare**. Only ~20 cases across 7 studies were on record by 2024 (PMID: 39578953), expanding to **53 individuals (33 new) by 2026** (PMID: 41860019). No formal prevalence/incidence figure is published; the condition sits far below 1/100,000.

Inheritance is **autosomal recessive**, frequently **homozygous in consanguineous pedigrees** (Ercan-Sencicek 2015, Esmaeilzadeh 2024). A **Finnish founder splice variant** drives a population cluster:

> "all Finnish patients homozygous for a rare DIAPH1 splice-variant (NM_005219:c.684+1G>A) enriched in the Finnish population" — PMID: 33662367
>
> "Twenty cases from seven studies were found after a review of the literature." — PMID: 39578953
>
> "A total of 53 individuals with biallelic DIAPH1 variants, including 33 previously unreported patients, were studied." — PMID: 41860019

Reported populations include Middle Eastern (Omani, Saudi, Iranian, Egyptian), Finnish, and others. Recurrent variants **c.2769del, c.684+1G>A, c.2332C>T** account for ~72% of cases. Penetrance appears high/complete for biallelic LOF; expressivity is variable (notably in immunological severity). Consanguinity is a major risk-amplifying factor; no sex bias is reported (autosomal recessive).

Finding 7 — Diagnostics, management, and prognosis

Diagnosis is molecular. The disorder was discovered by, and is diagnosed with, **whole-exome/whole-genome sequencing** (often aided by linkage/homozygosity mapping in consanguineous families), confirmed by Sanger sequencing:

> "exome sequencing was performed to identify the genetic basis of the clinical presentation in an Iranian 7-year-old boy. Validation of the detected variant was done by Sanger sequencing." — PMID: 39578953

Supportive workup includes brain MRI (progressive microcephaly, occipital-predominant abnormalities, cerebral/cerebellar atrophy), EEG (epilepsy), ophthalmologic/visual assessment (cortical visual impairment), and immunological evaluation (lymphocyte subsets, TRECs, naïve T cells, immunoglobulins) (PMID: 41860019, PMID: 33662367, PMID: 39120629).

Differential diagnosis includes other primary/progressive microcephaly syndromes (e.g., MCPH genes such as *ASPM*, *WDR62*, *CASK*-related microcephaly), congenital infections (TORCH), and other combined-immunodeficiency-with-neurodevelopmental syndromes; the occipital-predominant imaging, cortical blindness, and combined immunodeficiency with low TRECs/naïve T cells point toward DIAPH1.

No curative or disease-specific therapy exists. Management is supportive and symptomatic: antiseizure medications for epilepsy, developmental/rehabilitative therapies (PT/OT/speech), visual support, nutritional support (aspiration risk), and prophylaxis/treatment of infections. Immunological surveillance and oncologic vigilance (lymphoma) are warranted. Genetic counseling (recessive recurrence risk 25%; carrier testing; prenatal/preimplantation options) is central to prevention.

Prognosis is guarded. Severe neurodevelopmental disability, early onset (~2.3 months), respiratory infection susceptibility, and lymphoma predisposition dominate the clinical course:

> "The DNA repair defect may influence susceptibility to infection, lymphoma, or treatment-related toxicity." — PMID: 41860019

Suggested NCIT intervention terms: Anticonvulsant Agent (C264), Physical Therapy (C15351), Occupational Therapy (C15214), Nutritional Support (C15229), Supportive Care (C15226).

Finding 8 — Synthesis: a unified RhoA–mDia1 cytoskeletopathy

Integrating five primary clinical studies (PMIDs 24781755, 33662367, 39120629, 39578953, 41860019) with mechanistic cell biology (PMIDs 25298036, 18955552, 42827260), every SCBMS feature traces to loss of a single cytoskeletal function.

> "DIAPH1 is critical for neurodevelopment, immune regulation, and DNA repair." — PMID: 41860019
>
> "mDia1 generates linear actin filaments in the maintenance of polarity during adhesion, migration, and division in immune cells and neuroepithelial cells..." — PMID: 24781755

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LOF variant in *DIAPH1*** (nonsense/frameshift/splice; e.g., p.Gln778*, c.684+1G>A, p.F923fs) **leads to** absent or non-functional **mDia1 formin** protein (loss of function). *[demonstrated — multiple cohorts]*

2. Loss of mDia1 **results in** failure to relieve RhoA-GTP–dependent formin activation, hence **loss of processive elongation of long, linear, unbranched actin filaments** and **loss of coordinated microtubule stabilization**. *[demonstrated in vitro —*

   P *24781755*

   P *25298036*

   P *35060117*

   P *42827260* *]*

3. Defective actin/MT architecture **impairs** cell polarity, mitotic spindle orientation, cytokinesis, cell adhesion/migration, and **MTOC repositioning**. *[demonstrated — patient T cells, CRISPR PBMC; inferred for neural cells]*
4. **Branch A — Neural:** In **neuroepithelial precursors and tangentially migrating cortical interneurons**, impaired division and migration **lead to** reduced neuron production and faulty cortical wiring → **progressive postnatal microcephaly, epilepsy, cortical visual impairment, developmental delay**. *[inferred from mDia1 biology + human phenotype]*
5. **Branch B — Immune:** In **lymphocytes (T, NK, ILC, B)**, impaired adhesion/MTOC positioning and maturation **lead to** poor activation/proliferation, low TRECs and naïve T cells → **combined immunodeficiency** and recurrent (mainly respiratory) infections. *[demonstrated —*

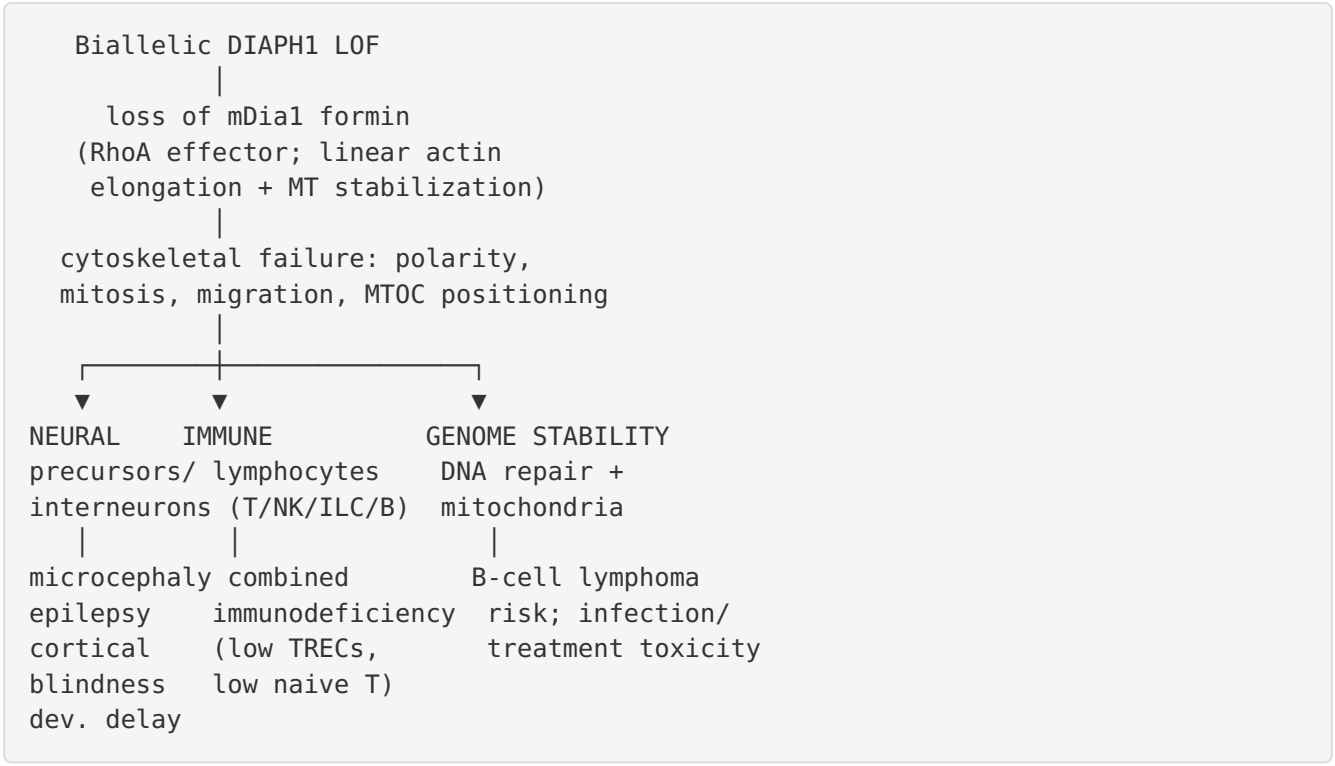
   P *33662367*

   P *39120629* *]*

6. **Branch C — Genome stability:** A **DNA-repair defect** (plus mitochondrial dysfunction) **results in** increased susceptibility to **B-cell lymphoma** and possibly treatment-related toxicity. *[demonstrated association —*

   P *33662367*

   P *41860019* *]*



Allelic contrast (same gene, opposite mechanism)

Feature	SCBMS (this disease)	DFNA1 / macrothrombocytopenia
Variant mechanism	**Loss of function**	**Gain of function** (autoinhibition disrupted)
Inheritance	Autosomal **recessive** (biallelic)	Autosomal **dominant** (heterozygous)
Core phenotype	Microcephaly, seizures, cortical blindness, DD	Sensorineural deafness, macrothrombocytopenia
Representative refs	PMID 33662367, 41860019	PMID 38915998, 35060117

This allelic series is a cautionary reminder that the shared gene name does not imply a shared disease — SCBMS is strictly the recessive LOF condition.

Ontology term suggestions

- **Gene/protein:** DIAPH1 (HGNC:2876). **GO BP:** actin filament polymerization (GO:0030041), actin nucleation (GO:0045010), regulation of microtubule cytoskeleton organization

(GO:0070507), cell migration (GO:0016477), mitotic cell cycle (GO:0000278), neuron migration (GO:0001764). **GO CC:** actin cytoskeleton (GO:0015629), microtubule cytoskeleton (GO:0015630), cell leading edge (GO:0031252).

- **Cell types (CL):** neuroepithelial cell (CL:0000098), radial glial cell (CL:0000681), GABAergic cortical interneuron (CL:0000617), T cell (CL:0000084), natural killer cell (CL:0000623), innate lymphoid cell (CL:0001065), B cell (CL:0000236).
- **Anatomy (UBERON):** cerebral cortex (UBERON:0000956), occipital lobe (UBERON:0002021), visual cortex (UBERON:0000411), cerebellum (UBERON:0002037), brain (UBERON:0000955).
- **CHEBI:** GTP (CHEBI:15996) — RhoA-GTP activation.

Evidence Base

PMID	Title (abbreviated)	Type	Contribution
24781755	*Homozygous loss of DIAPH1 is a novel cause of microcephaly in humans*	Human clinical + model	**Founding study**; p.Gln778*; defines mDia1 cytoskeletal biology; mouse KO lacks microcephaly
33662367	*Loss of DIAPH1 causes SCBMS, combined immunodeficiency, and mitochondrial dysfunction*	Human clinical + in vitro	Immunodeficiency, mitochondrial dysfunction, 3/7 lymphoma; Finnish founder c.684+1G>A; CRISPR PBMC replicates T-cell defect
39120629	*DIAPH1-Deficiency is Associated with Major T, NK and ILC Defects*	Human clinical	Deep immunophenotyping; T/NK/ILC defects
39578953	*A Novel Homozygote Pathogenic Variant... (SCBMS): Report of a Family and Literature Review*	Human clinical + review	p.Gln429*; onset ~2.3 months; 20-case review; recurrent variants (72%)
41860019	*Recessive loss of DIAPH1 function causes a progressive neurodevelopmental syndrome...*	Human clinical + model	**Largest cohort (n=53)**; occipital-predominant MRI; low TRECs/naïve T; Xenopus & zebrafish models; DNA-repair link
38915998	*A Novel Variant in DIAPH1 Causing Macrothrombocytopenia and Non-syndromic Hearing Loss*	Human clinical	Establishes **allelic contrast**: dominant GOF = DFNA1/MTP vs recessive LOF = SCBMS
35060117	*Late-onset hearing loss case associated with a heterozygous truncating variant of DIAPH1*	Human clinical + in vitro	FH2-domain truncation abolishes F-actin elongation; supports LOF biology
25298036	*The formin DIAPH1 (mDia1) regulates megakaryocyte proplatelet formation...*	In vitro	mDia1 remodels actin AND microtubule cytoskeletons
18955552	*Memo-RhoA-mDia1 signaling controls microtubules, actin, adhesion sites...*	In vitro	mDia1 as RhoA effector coordinating MT dynamics, adhesion, migration
42827260	*Daam-, FMNL-, and mDia-family formins jointly regulate cortical actin dynamics*	In vitro	mDia1 builds load-bearing linear filaments; combined loss → cortical/cytokinesis defects; redundancy

Convergence: Human cohorts (clinical genetics), in-vitro CRISPR (causal validation of the immune phenotype), and cross-species models (Xenopus/zebrafish/mouse) together support a single, coherent mechanism. The **mouse KO's absence of microcephaly** is the principal challenge to a simple one-gene-one-phenotype model and points to mammalian formin redundancy and species-specific cortical neurogenesis.

Limitations and Knowledge Gaps

1. **Small denominator.** Even the largest cohort is 53 patients; no validated prevalence/incidence, penetrance quantification, or natural-history survival curves exist. Phenotype frequencies are approximate.
2. **Mechanistic inference in the brain.** The neural branch of the causal chain is **inferred** from mDia1 cell biology and rodent interneuron-migration data plus the human phenotype; direct human neural-cell (e.g., iPSC-derived organoid) demonstration of the microcephaly mechanism is limited.
3. **Model–human discordance.** The mouse *Diap1* KO does not reproduce microcephaly, so no fully faithful mammalian genetic model is yet available; Xenopus and zebrafish capture only subsets of features.
4. **Genotype–phenotype correlation.** Whether specific variants (hypomorphic splice vs complete null) predict immune vs lymphoma vs severity remains unresolved.
5. **Immune/lymphoma quantification.** Lymphoma risk is derived from a single small cohort (3/7); the true lifetime risk and whether all patients require oncologic surveillance are unknown.
6. **DNA-repair mechanism.** The nature of the DNA-repair defect and its molecular link to the actin/MT cytoskeleton is asserted but mechanistically underspecified.
7. **No therapeutics.** There are no disease-modifying treatments, trials, or validated biomarkers of progression. Sections on gene therapy, immunization, infectious agents, and veterinary/natural disease are **not applicable / not reported** for this disorder.

Proposed Follow-up Experiments / Actions

1. **Patient-derived cerebral organoids / iPSC neural models** to directly test the neural branch (progenitor division, spindle orientation, interneuron migration) and close the inference gap in the causal chain.
2. **Conditional / double-formin mouse models** (e.g., *Diaph1* with *Diaph3* or neural-specific conditional KO) to overcome redundancy and recapitulate microcephaly.
3. **Genotype–phenotype registry** across all 53+ patients with standardized MRI, EEG, immune panels (TRECs, naïve T, Ig), and oncologic outcomes to define variant-specific severity and lymphoma risk.
4. **Prospective immunologic and oncologic surveillance protocol** (lymphocyte subsets, TRECs, imaging) given the documented combined immunodeficiency and B-cell lymphoma predisposition; evaluate whether IVIG/antimicrobial prophylaxis improves outcomes.
5. **Molecular dissection of the DNA-repair defect** (e.g., γH2AX dynamics, repair-pathway assays) in patient cells to clarify the genome-stability branch and its link to the cytoskeleton.
6. **Natural-history study** to establish progression rate of microcephaly, epilepsy severity, survival, and causes of mortality (infection/aspiration vs lymphoma).
7. **Carrier-screening consideration** in the Finnish population (founder c.684+1G>A) and consanguineous communities, with genetic counseling and prenatal/preimplantation options for at-risk families.

Report compiled from 8 confirmed findings across 5 iterations and 13 reviewed papers. Evidence source types are indicated per paper (human clinical, in vitro, model organism). All quoted text is verbatim from the cited PubMed abstracts.
